

23CSE352-BIG DATA ANALYTICS

PROJECT REVIEW 1- 15 Marks

Important Instructions

- **The project must use both Hadoop MapReduce and Apache Hive.**
- **The dataset should be a real-world dataset and not a manually created sample dataset.**
- **Ensure that the project demonstrates a meaningful real-world application.**

The objective of Project Review – I is to evaluate understanding and practical implementation of Big Data technologies using **Hadoop MapReduce** and **Apache Hive** on a **real-world dataset**. Students are expected to demonstrate the complete data processing pipeline from data ingestion to analysis.

Project Requirements:

Each team must select one real-world dataset from any application domain. The selected dataset should contain sufficient records (preferably 5,000 records or more) to demonstrate Hadoop processing.

Implementation Requirements

Dataset Preparation

- Select a real-world dataset.
- Explain the application domain.
- Describe each attribute in the dataset.
- Perform necessary preprocessing (if required).
- Upload the dataset into HDFS.

Hadoop MapReduce

Implement **at least one meaningful MapReduce program** based on the selected application.

Examples include:

- Sales Analysis
- Maximum/Minimum Analysis
- Average Calculation
- Category-wise Statistics

- Frequency Analysis
- Customer/Product Analysis
- Log File Analysis
- Performance Analysis

Apache Hive

Load the same dataset into Hive and perform data analysis using SQL-like Hive Queries.

Students should demonstrate **at least 10 meaningful Hive queries**, including:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- COUNT
- SUM
- AVG
- MAX
- MIN
- JOIN (if multiple tables are used)

Queries should address real-world business or analytical questions related to the dataset.

Demonstration During Review

Each team should demonstrate:

- Problem statement and Motivation
- Dataset introduction
- HDFS storage
- MapReduce execution
- Hive table creation
- Hive query execution
- Results and interpretation
- Real-world application of the project

Every team member must actively participate in the presentation and answer questions related to the implementation.

Document Requirements

Each team should submit Project Report (PDF)

Students should clearly explain:

1. Introduction
2. Problem Statement
3. Motivation
4. Dataset (or dataset source link)
5. Architecture
6. MapReduce Problem
7. Mapper logic explanation
8. MapReduce source code
9. Reducer Logic
10. Reducer code
11. Input and Output
12. Screenshots of execution
13. Output screenshots
14. Hive analysis questions
15. Each query executions
16. Screenshot Query result
17. Conclusion

Evaluation Rubrics (15 Marks)

Criteria	Marks
Problem identification and relevance of the real-world application	2
Dataset selection, preprocessing, and HDFS storage	2
Hadoop MapReduce implementation and successful execution	4
Hive table creation and execution of meaningful analytical queries	4
Project demonstration, explanation, interpretation of results, and individual participation	3
Total	15 marks

Sample Applications

No. Mini Project Title

- 1 Customer Churn Prediction
- 2 Credit Risk Prediction for Loan Approval using Big Data Analytics
- 3 Fraud Detection in Financial Transactions
- 4 Stock Market Trend Analysis
- 5 Healthcare Patient Risk Prediction
- 6 Disease Pattern Analysis from Healthcare Big Data
- 7 Medical Insurance Claim Prediction using Big Data Analytics
- 8 Sentiment Analysis of Social Media Data
- 9 Fake News Detection using Big Data Text Analytics
- 10 Movie Recommendation System
- 11 E-Commerce Product Recommendation using Big Data Analytics
- 12 Customer Segmentation
- 13 Market Basket Analysis for Retail Transactions
- 14 Sales Prediction and Forecasting
- 15 Retail Customer Behaviour Analysis using Big Data Processing
- 16 Student Performance Prediction
- 17 Learning Analytics System for Online Education Platforms
- 18 Traffic Congestion Prediction using Big Data Analytics
- 19 Vehicle Accident Analysis
- 20 Smart City Data Analytics
- 21 Weather Pattern Analysis and Prediction
- 22 Air Quality Monitoring and Prediction using Big Data Analytics
- 23 Electricity Consumption Prediction
- 24 Electricity Theft Detection using Big Data Analytics
- 25 Cyber Security Intrusion Detection

No. Mini Project Title

- 26 **Network Traffic Analysis**
- 27 **Web Server Log Analysis**
- 28 **Search Query Pattern Analysis**
- 29 **Online Payment Transaction Analysis**
- 30 **Supply Chain Demand Prediction**
- 31 **Agricultural Crop Yield Prediction**
- 32 **Smart Farming Data Analysis**
- 33 **Airline Flight Delay Prediction**
- 34 **Employee Attrition Prediction**
- 35 **Hotel Booking Cancellation**